Design of Test Cases and Maintenance Plan for a Hospital Management System

Aim:

To design and develop comprehensive test cases and prepare an effective maintenance plan for a Hospital Management System to ensure its functionality, reliability, security, and continuous improvement.

Objective:

The objective of this project is to design effective test cases and develop a comprehensive maintenance plan for the Hospital Management System to ensure that all its modules function accurately, efficiently, and securely. The project aims to verify the performance of key functionalities such as patient registration, appointment scheduling, medical records management, laboratory services, pharmacy, billing, and user authentication through systematic testing

Test Cases:

Testing ensures that every module of the Hospital Management System (HMS) functions correctly, securely, and efficiently.

Test Case ID	Module	Test Scenario	Input	Expected Result	Status
TC-01	Patient registration	Register a new patient	Valid patient details	Patient registered successfully and generated unique ID	Pass
TC-02	Patient login	Login with valid credentials	Username and password	Login successfully	Pass
TC-03	Appointment	Book an appointment	Select doctor, date and time	Appointment conformed	Pass
TC-04	Medical records	View patient medical history	Valid patient ID	Complete medical records displayed	Pass
TC-05	Pharmacy	Issue prescribed medicines	Valid prescription	Medicines issued and inventory updated	

Types of Testing:

- **Unit Testing -** Tests individual modules such as Registration, Billing, Pharmacy, and Laboratory independently.
- **Integration Testing -** Verifies communication between modules such as Appointment ↔ Doctor Schedule and Billing ↔ Pharmacy.
- **System Testing -** Ensures the complete Hospital Management System functions as expected.
- **User Acceptance Testing -** Conducted by doctors, nurses, receptionists, and administrators to verify the system meets hospital requirements.
- **Performance Testing -** Checks system performance when many users access it simultaneously.
- **Security Testing -** Ensures patient information is protected using authentication, authorization, and encryption.

Maintenance Plan:

After deployment, the Hospital Management System requires regular maintenance to ensure smooth operation, security, and performance.

Types of Maintenance:

1. Corrective Maintenance:

Purpose:

- Fix software bugs and errors reported by users.

Examples:

- Resolve appointment booking failures.
- Correct billing calculation errors.
- Fix login issues.

2. Adaptive Maintenance:

Purpose:

- Modify the software to meet new business or regulatory requirements.

Examples:

- Update the system according to new healthcare regulations.
- Add support for new payment gateways.
- Integrate new laboratory equipment.

3. Perfective Maintenance:

Purpose:

- Improve system performance and add new features.

Examples:

- Faster patient search.
- Improved user interface.
- Add telemedicine functionality.
- Enable online appointment reminders.

4. Preventive Maintenance:

Purpose:

- Prevent future issues and improve system reliability.

Examples:

- Regular database optimization.
- Software updates and patches.
- Data backup and recovery testing.
- Security vulnerability scanning.

Expected Outcomes:

Reliable and error-free hospital operations.

- Secure management of patient records.
- Improved system performance and availability.
- Faster issue resolution.
- Compliance with healthcare regulations.
- Increased patient and staff satisfaction.
- Continuous enhancement of system features.

Conclusion:

A well-designed testing process ensures that every module of the Hospital Management System operates accurately and securely before deployment. A structured maintenance plan helps keep the system reliable, efficient, and up to date by addressing bugs, improving performance, applying security updates, and introducing new features. Together, testing and maintenance contribute to high-quality healthcare services and uninterrupted hospital operations.